

# Dirac Delta Function

## Important Properties

$$1. \int_{-\infty}^{\infty} f(x) \delta(x - x_0) dx = f(x_0)$$

$$\text{Example: } \int_{-\infty}^{\infty} e^{-\alpha x^2} \delta(x - 1) dx = e^{-\alpha}$$

$$2. \delta(x - x_0) = 0 \qquad \delta(x - x_0) = \infty$$

$$\text{If } x \neq x_0$$

$$\text{If } x = x_0$$

$$3. \delta(ax) = \frac{1}{|a|} \delta(x)$$

$$4. \delta(x^2 - a^2) = \frac{1}{2|a|} [\delta(x + a) + \delta(x - a)]$$

$$5. \delta(g(x)) = \sum_i \frac{\delta(x-x_i)}{|g'(x_i)|}$$

$x_i \rightarrow \text{zero's}$  By putting  $g(x_i) = 0$

**Example:**  $\delta(x^2 - a^2)$

$$g(x) = x^2 - a^2$$

$$g'(x) = 2x$$

$$g(x) = 0 = x^2 - a^2 = 0$$

$$x_i = a, -a$$

$$x_i = a, x_i = -a$$

$$\therefore \delta(x^2 - a^2) = \frac{\delta(x-a)}{|a|} + \frac{\delta(x+a)}{|-a|} = \frac{1}{2a} [\delta(x-a) + \delta(x+a)]$$

**Example:**  $\delta(x^2 + x - 2)$

$$g(x) = x^2 + x - 2 \quad g'(x) = 2x + 1$$

$$g(x) = 0$$

$$x = \frac{-1 \pm \sqrt{1+8}}{2} = \frac{-1 \pm 3}{2} = 1, -2$$

$$\therefore \delta(x^2 + x - 2) = \frac{\delta(x-1)}{|3|} + \frac{\delta(x+2)}{|-3|} = \frac{\delta(x-1)}{3} + \frac{\delta(x+2)}{3}$$

$$\text{So, } \int_{-\infty}^{\infty} \delta(x^2 + x - 2) dx = \frac{1}{3} \int_{-\infty}^{\infty} [\delta(x-1) + \delta(x+2)] dx$$

$$= \frac{1}{3} [1 + 1] = 2/3$$

$$6. \int f(x) \delta^{(n)}(x) dx = - \int \frac{\partial f}{\partial x} \delta^{(n-1)}(x) dx$$

Where,  $\delta^{(n)}(x)$  is nth derivative of the delta function.

Example:  $f(x) = x\delta(x)$

$$\int xg(x)\delta'(x)dx = -\int [xg'(x) + g(x)]\delta(x)dx$$

$$= \int xg'(x)\delta(x)dx - \int g(x)\delta(x)dx$$

$$\therefore \int xg(x)\delta'(x) = \int (-g(x))\delta(x)dx$$

$$x\delta(x)\delta'(x) = -\delta(x) \cdot g(x)$$

$$x\delta'(x) = -\delta(x)$$

$$x^n\delta^n(x) = (-1)^n n!\delta(x)$$

$$x^2\delta''(x) = (-1)^2 2!\delta(x)$$

$$x^2\delta''(x) = 2 \cdot \delta(x)$$

$$7. \delta'(-x) = -\delta'(x)$$

$$8. \int_{-\infty}^{\infty} f(x)\delta'(x-a)dx = -f'(a)$$

$$f(x)\delta(x-a) \int_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f'(x)\delta(x-a)dx = -f'(a)$$

$$9. \int_{-1}^1 \delta\left(\frac{1}{x}\right)dx = 0 \Rightarrow \delta(f(x)) = \frac{\delta(x-x_i)}{|f'(x_i)|}$$

$$\therefore \frac{1}{32} = 0$$

$$\Rightarrow x = \pm\infty = \frac{\delta(x-\infty)}{\frac{1}{x^2}} + \delta\left(\frac{x+\infty}{\frac{1}{x^2}}\right)$$

$$= \int_{-1}^1 \left( \frac{x^2\delta(x-\infty)}{1} + \frac{x^2\delta(x+\infty)}{1} \right) = 0 \text{ The delta functions are not in range.}$$

$$10. \text{ If } \int_{-\infty}^{\infty} f(x)\delta(x)dx = f(0)$$

$f(x)$  is a well-behaved function and the dimension of  $x$  is momentum.

**Question:** What is dimension of  $\delta(x)$ ?

- (a) Dimension less (b) Momentum  
(c) Momentum (d) Momentum

**Solution:** Dimensionally  $f(x)\delta(x-x_0)dx = f(0)$

$\delta(x)dx$  should be dimensionless.

$$\delta(x) = [p]^{-1}$$

$$11. \int_{-\infty}^{\infty} (x^2 + 1)\delta(x^2 - 3x + 2)dx$$

$$\int_{-\infty}^{\infty} (x^2 + 1)\delta[(x-2)(x-1)]dx$$

$$= \int_{-\infty}^{\infty} (x^2 + 1) \cdot \left( \frac{\delta(x-2)}{|2x-3|} + \frac{\delta(x-1)}{|2x-3|} \right) dx$$

$$= \int_{-\infty}^{\infty} (x^2 + 1) \left( \frac{\delta(x-2)}{1} + \frac{\delta(x-1)}{1} \right) dx = (2^2 + 1) + (1^2 + 1) = 5 + 2 = 7$$

$$12. \pi \int_{-\infty}^{\infty} e^{-|x|} \delta(\sin \pi x) dx$$

$$f(x) = \sin \pi x; \pi x = n\pi$$

$$f'(x) = \pi \cos \pi x$$

$$\therefore \pi e^{-|x|} \left( \frac{\delta(x-n)}{\pi \cos(n\pi)} + \dots \right) = 2e^{-|x|} \left( \frac{\delta(x-n)}{\pi} \right) = \frac{2e^{-|x|}}{\pi} \cdot (e^{-1} + e^{-2} + \dots)$$

$$= \frac{2e^{-1}}{\pi} \cdot \frac{e^{-1}}{(1-e^{-1})} \Rightarrow \frac{2e^{-1}}{\pi} \cdot \frac{e}{(e-1)} = \frac{2e^{-1}}{\pi} \cdot \frac{e}{(e-1)} = \frac{2e^{-1}}{\pi} \cdot \frac{e}{(e-1)}$$

$$\text{or, } \delta(\sin \pi x) = \dots + \frac{\delta(x+2)}{|\pi \cos(-2\pi)|} + \frac{\delta(x+1)}{|\pi \cos(-\pi)|} + \frac{\delta(x)}{|\pi \cos 0|} + \frac{\delta(x-1)}{|\pi \cos(\pi)|} + \frac{\delta(x-2)}{|\pi \cos 2\pi|} + \dots$$

$$= \int_{-\infty}^{\infty} e^{-|x|} (\delta \dots)$$

$$= 1 + 2(e^{-1} + e^{-2} + \dots) = 1 + 2(e^{-1} + e^{-2} + \dots) = 1 + \frac{2}{e} \cdot \frac{e}{(e-1)} \Rightarrow \left( \frac{e+1}{e-1} \right)$$

Example:  $\int_{-\infty}^{\infty} dx \quad \delta(x^2 - \pi^2) \cos x$

$$\delta(x^2 - \pi^2) = \delta[(x - \pi)(x + \pi)] = \frac{\delta(x - \pi)}{|2\pi|} + \frac{\delta(x + \pi)}{|-2\pi|}$$

$$\therefore \frac{1}{2\pi} [\cos \pi + \cos(-\pi)] = -\frac{1}{\pi}$$

Example:  $\int_0^3 t^2 \delta(3t - 1) dt = \frac{4}{3}$

Example:  $\int_{-\pi/2}^{\pi/2} dx \int_{-1}^1 dy \quad \delta(\sin 2\pi) \delta(x - y)$

$$= \int_{-\pi/2}^{+\pi/2} dx \quad \delta(\sin(2x)) \int_{-1}^1 dy \delta(x - y)$$

From here,  $y = x, -1 < y < 1$

$$\pi/2 = +1$$

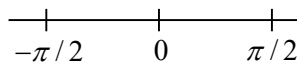
$$\int \delta(\sin 2x) dx$$

$$-\pi/2 = -1$$

$$f(x) = \sin 2x = 0$$

$$f'(x) = 2 \cos(2x)$$

$$x = \frac{n\pi}{2}$$



$$= \int_{-\pi/2}^{\pi/2} dx \delta(\sin 2x) = \frac{1}{2} = \int_{-\pi/2}^{\pi/2} dx \frac{\delta(x)}{|2 \cos(0)|}$$

Example:  $x\delta'(x) = -\delta(x)$

A differential equation containing Delta function

$$\# \frac{d^2 f(x)}{dx^2} - \alpha \delta(x) f(x) = \beta f(x)$$

$$V = \alpha \delta(x)$$

$$\Delta \left( \frac{\partial f}{\partial x} \right) = ??$$

$$\therefore \int_{-\epsilon}^{+\epsilon} \frac{\partial^2 f}{\partial x^2} \partial x - \alpha \int_{-\epsilon}^{+\epsilon} \delta(x) f(x) \partial x = \beta \int_{-\epsilon}^{+\epsilon} f(x)$$

$$\Rightarrow \left. \frac{\partial f}{\partial x} \right|_{-\epsilon}^{+\epsilon} - \alpha f(0) = \beta(0)$$

$$\Delta \left[ \frac{\partial f}{\partial x} \right] = \alpha f(0)$$

$$f = Ae^{vx}; x < 0 = Be^{-vx}; x > 0$$

$A = B$  By boundary condition

$$\Delta \left[ \frac{\partial f}{\partial x} \right] = \alpha A$$

$$Ave^{v \cdot 0} - A(-v)e^{-v \cdot 0} - \alpha A = 0$$

$$Ave^{v \cdot 0} - A(-v)e^{-v \cdot 0} - \alpha A = 0$$

$$v = \alpha / 2$$

**Example:**  $\delta(x - x_0) = \begin{cases} 0 & x \neq x_0 \\ \infty & x = x_0 \end{cases}$

One dimensional dirac-delta function

$$\delta^2(x, y) = \begin{cases} 0 & x^2 + y^2 \neq 0 \\ \infty & x^2 + y^2 = 0 \end{cases}$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta^2(x, y) dx dy = 1$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(x - x_0) \delta(y - y_0) dx dy = 1$$

$$\delta^2(ax, by) = \frac{1}{|ab|} \delta^2(x, y)$$

Polar Coordinates

$$\delta^2(x, y) = \left( \frac{\delta(r)}{\pi r} \right)$$

Three Dimension

$$\delta^3(x, y, z) = \delta(x) \delta(y) \delta(z) = \delta^3(x)$$

$$= \begin{cases} 0 & x^2 + y^2 + z^2 \neq 0 \\ \infty & x^2 + y^2 + z^2 = 0 \end{cases}$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta^3(x, y, z) dx dy dz = 1$$

Cylindrical Coordinate

$$\delta^3(r, \phi, z) = \frac{\delta(r) \delta(z)}{\pi r}$$

Spherical Polar Coordinate

$$\delta^3(r, \theta, \phi) = \frac{\delta(r)}{2\pi r^2}$$

Series expansion cylindrical coordinate

$$\delta^3(r_1 - r_2) = \frac{1}{r_1} \delta(r_1 - r_2) \delta(\phi_1 - \phi_2) \delta(z_1 - z_2)$$

Series expansion spherical polar coordinate

$$\delta^3(r - r_0) = \frac{1}{r^2 \sin \theta} \delta(r - r_0) \delta(\theta - \theta_0) \delta(\phi - \phi_0)$$

Azimuthal symmetry

$$\int_0^{2\pi} \pi^2 \sin \theta d\phi = 2\pi r^2 \sin \theta$$

$$\delta(r - r_0) = \frac{1}{2\pi r^2 \sin \theta} \delta(r - r_0) \delta(\theta - \theta_0)$$

$\theta$  and  $\phi$  both symmetrical

$$\int_0^\pi 2\pi r^2 \sin \theta d\theta = 4\pi r^2$$

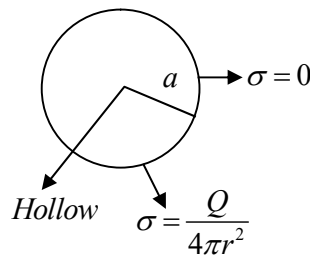
$$\delta^3(r - r_0) = \frac{1}{4\pi r^2} \delta(r - r_0)$$

**Physical Example:**

Choose density of a spherical shell

$$\sigma(r) = \frac{Q}{4\pi r^2} \delta(r - a)$$

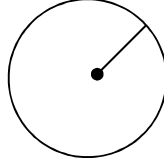
Ring



$$\lambda(r) = \frac{Q\delta(r-0)}{2\pi r}$$

For point charge

$$q = \frac{q_0\delta(r-r_0)}{4\pi r^2}$$



$$\int q_0 \frac{\delta(r-r_0)}{4\pi r^2} \cdot 4\pi r^2 dr = q_0$$